Process Analysis: Delivery Reliability Monitoring

Prepared by Bimal K Augustine, Business Analyst (portfolio case study) · Version 1.0 · 25 October 2026 · Final

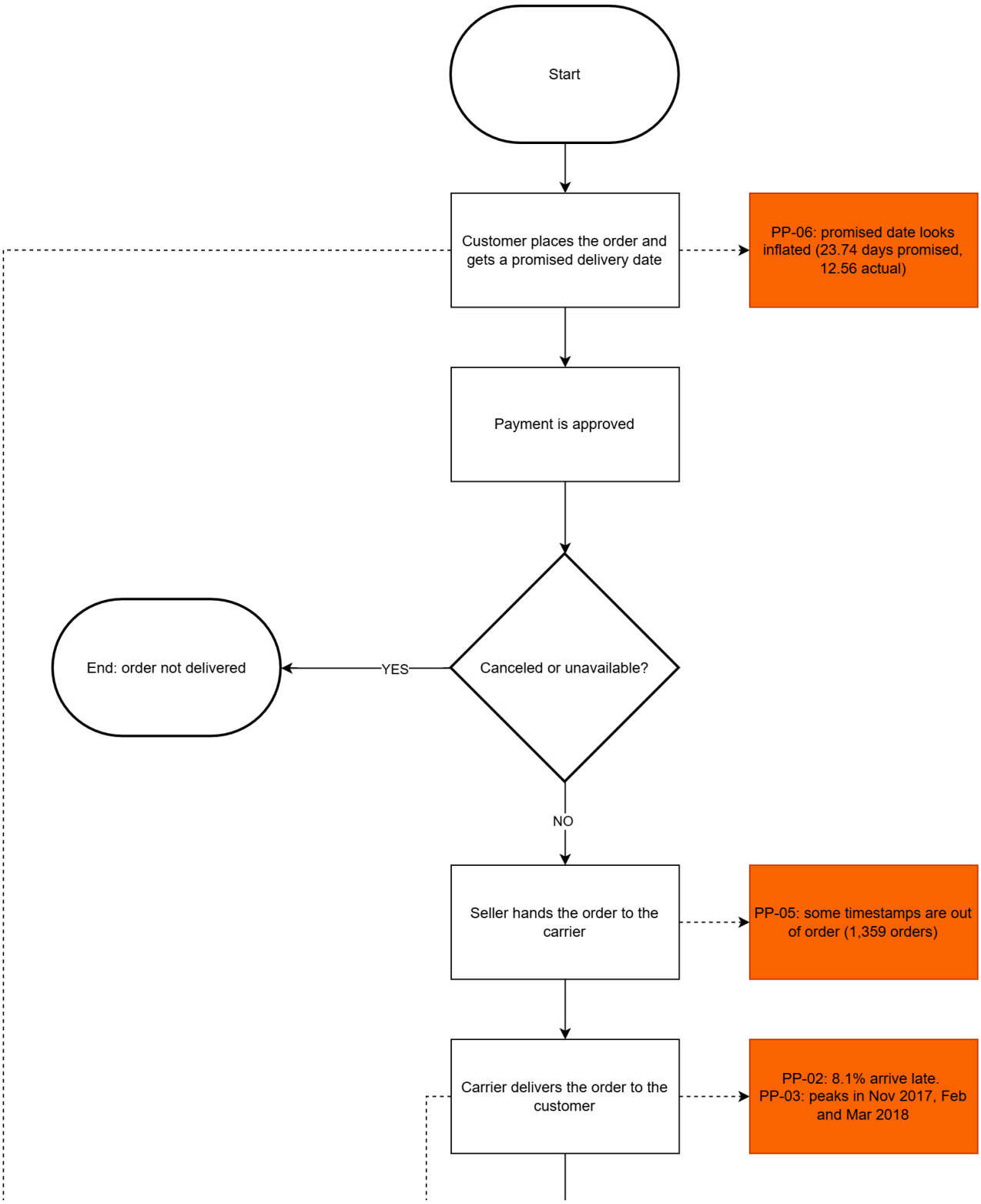
Today, the delivery date is recorded but nobody compares it with the date promised to the customer. This document maps the current process, lists its problems, and proposes a monthly review where each problem goes to the person who owns it.

1. AS-IS Process

An order passes through five recorded steps, and none of them compares the delivery date with the date promised at purchase.

Step	What happens	Data that records it
1	Customer places the order and is given a promised delivery date	order_purchase_timestamp, order_estimated_delivery_date
2	Payment is approved	order_approved_at
3	Seller hands the order to the carrier (seller has a shipping deadline)	order_delivered_carrier_date, shipping_limit_date
4	Carrier delivers the order to the customer	order_delivered_customer_date
5	Customer receives a survey and may leave a 1 to 5 score	review_creation_date, review_answer_timestamp, review_score
Exception	Order is canceled or unavailable and never reaches delivery	order_status

AS-IS process: order journey and pain points



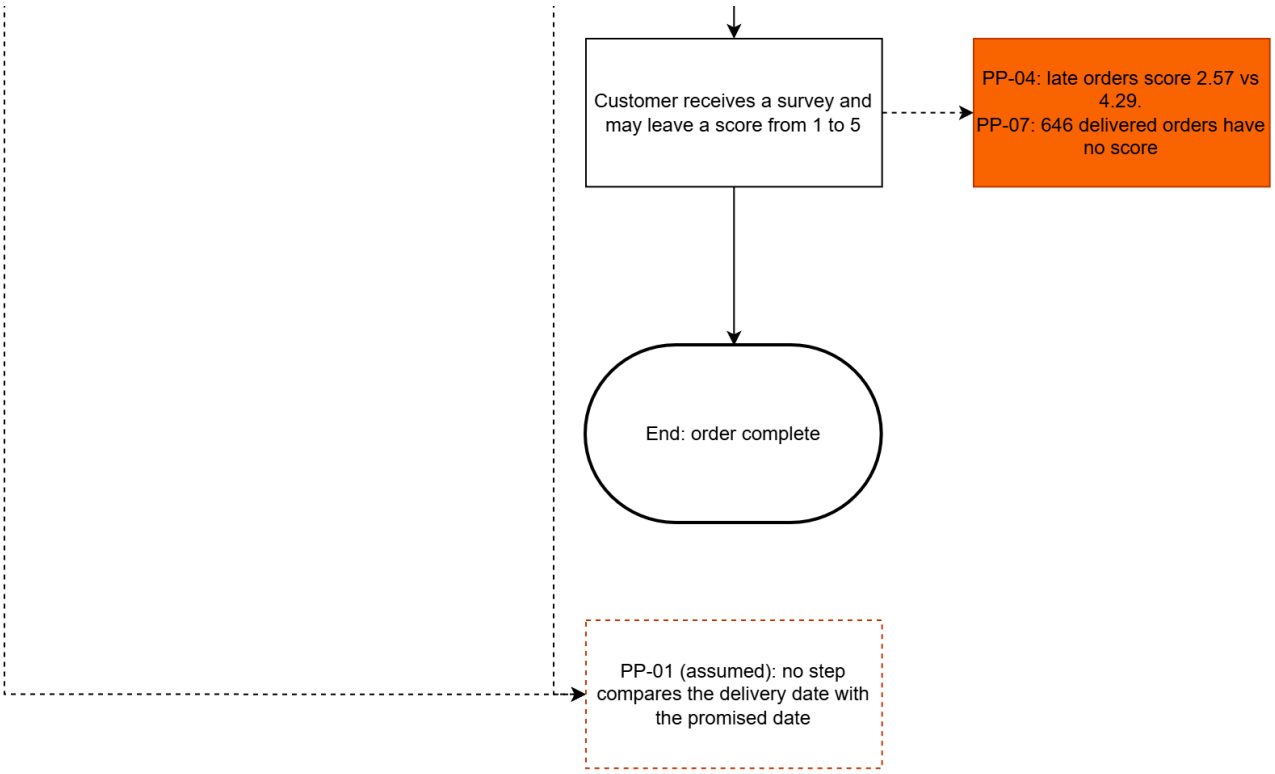


Figure 1. AS-IS order journey (01_as_is_process.png, drawn in draw.io). Dashed lines show where each pain point appears.

2. Pain Points and Gaps

Seven problems stand in the way of reliable delivery. Six are measured in the data, and one (PP-01) is the scenario assumed in the BRD.

ID	Pain point	Evidence	Gap closed by
PP-01	No routine check of delivery date against the promise	Scenario assumption from the BRD, not measured	BR-01, BR-10
PP-02	Late deliveries	7,826 of 96,470 delivered orders (8.1%) arrive after the promised date	BR-01, BR-02
PP-03	Lateness spikes in particular months	March 2018 21.4%, February 2018 16.0%, November 2017 14.3%	BR-03

ID	Pain point	Evidence	Gap closed by
PP-04	Late orders earn low ratings	Average score 2.57 for late orders against 4.29 on time; 2.32 for orders 3 to 7 days late	BR-06
PP-05	Stage timestamps are sometimes out of order	1,359 orders show carrier pickup before payment approval	BRU-01 (use only purchase, delivery and promised dates)
PP-06	The promised date looks padded	Orders are promised 23.74 days ahead on average but arrive in 12.56 days on average	BR-10
PP-07	Satisfaction is only partly visible	768 orders have no review; 646 delivered orders have no score	BR-06, BR-10

Not covered here: 1,234 orders were canceled (625) or unavailable (609). These orders never reached delivery, so they are outside the scope of this project (see the BRD).

3. TO-BE Process

A monthly loop compares delivery with the promise and sends each problem to the role that owns it. The 5% target is assumed.

What changes from the AS-IS	Pain point addressed	Requirement
Late rate and delivery time are reported every month with fixed definitions	PP-01, PP-02	BR-01, BR-02, BR-10
Peaks are visible by month, state, seller and category	PP-03	BR-03, BR-04, BR-05, BR-07
Review scores are shown by how late an order was	PP-04	BR-06
Data checks run before every refresh	PP-05, PP-07	BR-10

What changes from the AS-IS	Pain point addressed	Requirement
Each missed target triggers named actions with named owners	PP-02, PP-03	RACI activities 5 to 7

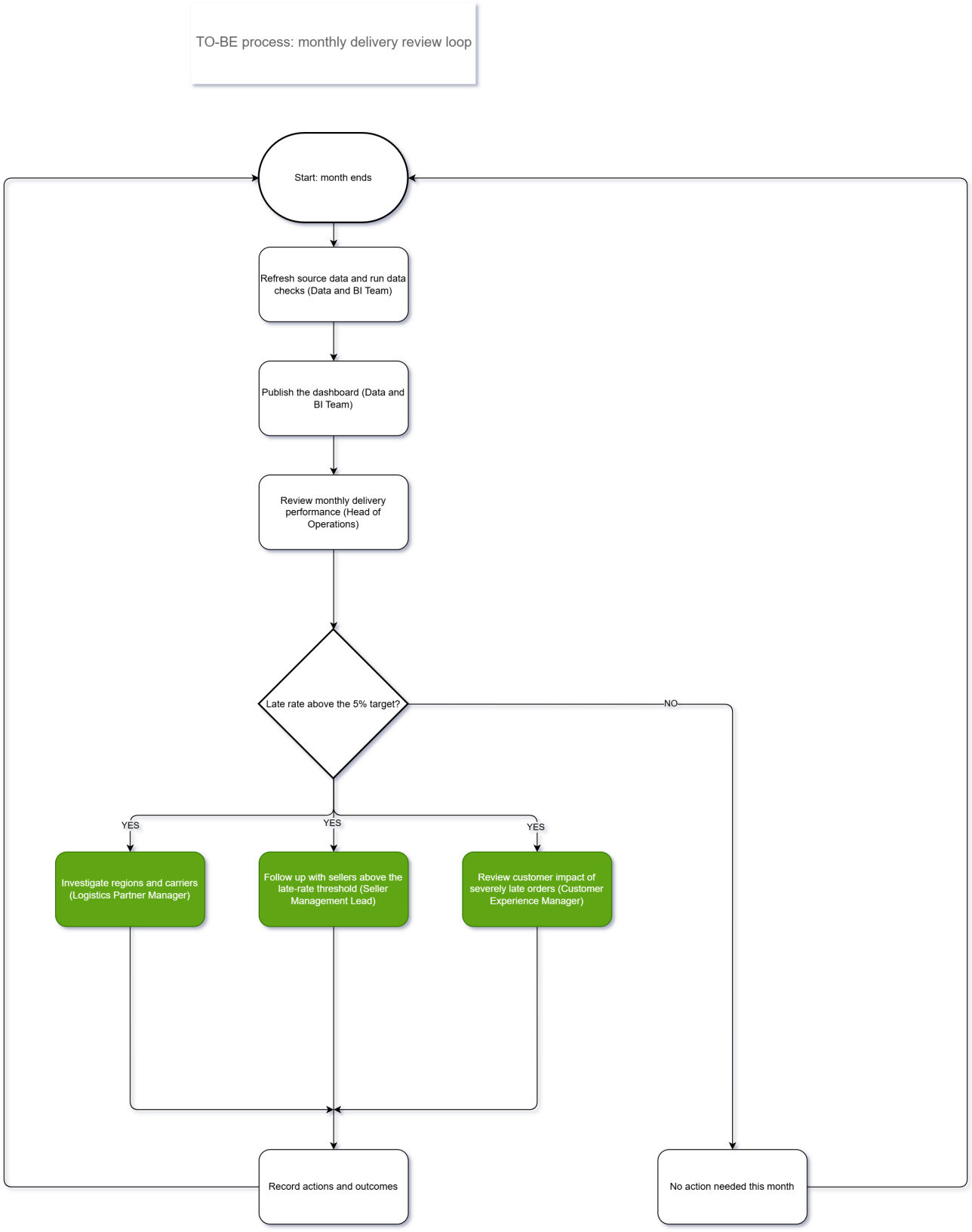
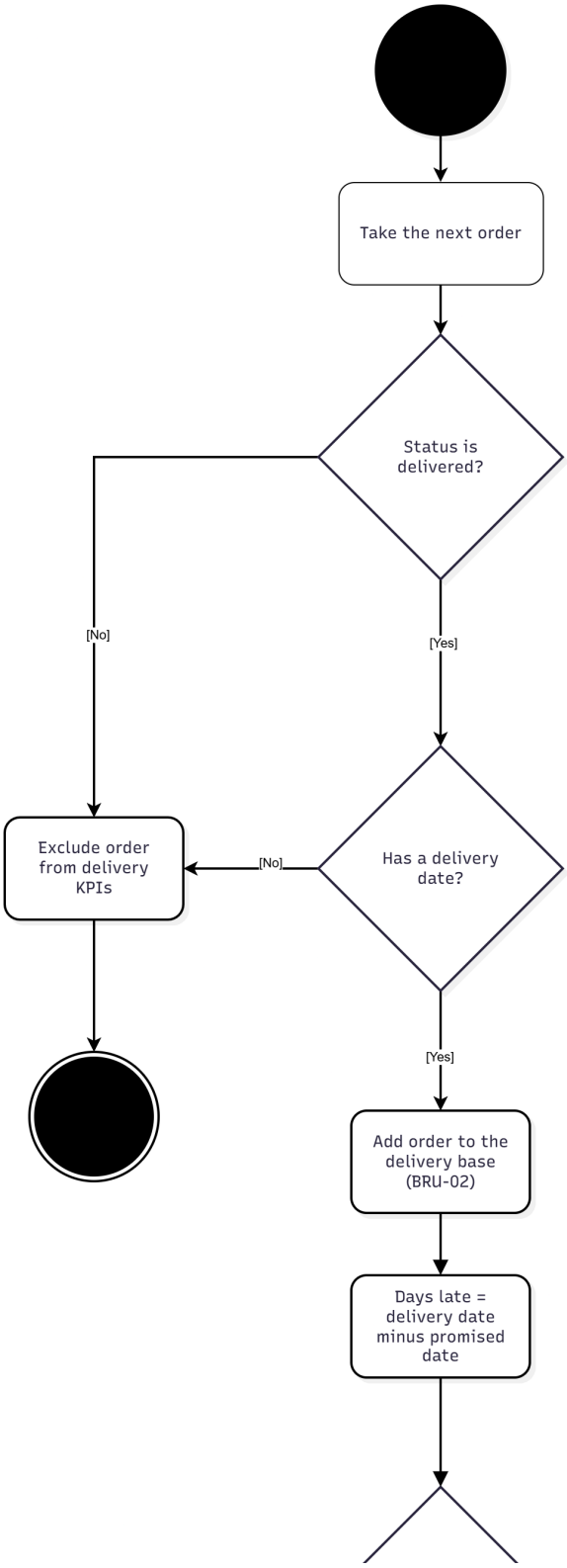


Figure 2. TO-BE monthly review loop (02_to_be_process.png, drawn in draw.io). Only a missed target sends work to the three owners; otherwise the loop repeats next month.

4. How an Order Is Counted as Late

The monthly review depends on one rule: which orders count, and when an order is late. Only delivered orders that have a delivery date count. An order is late if it arrives after the promised date. Late orders are then sorted into delay buckets (1-3, 3-7, 7-14 and over 14 days late), because the review score drops more the later an order is.


Take the next order
Status is delivered?
[No]
[Yes]
Has a delivery date?
[No]
Exclude order from delivery KPIs
[Yes]
Add order to the delivery base (BRU-02)
Days late = delivery date minus promised date

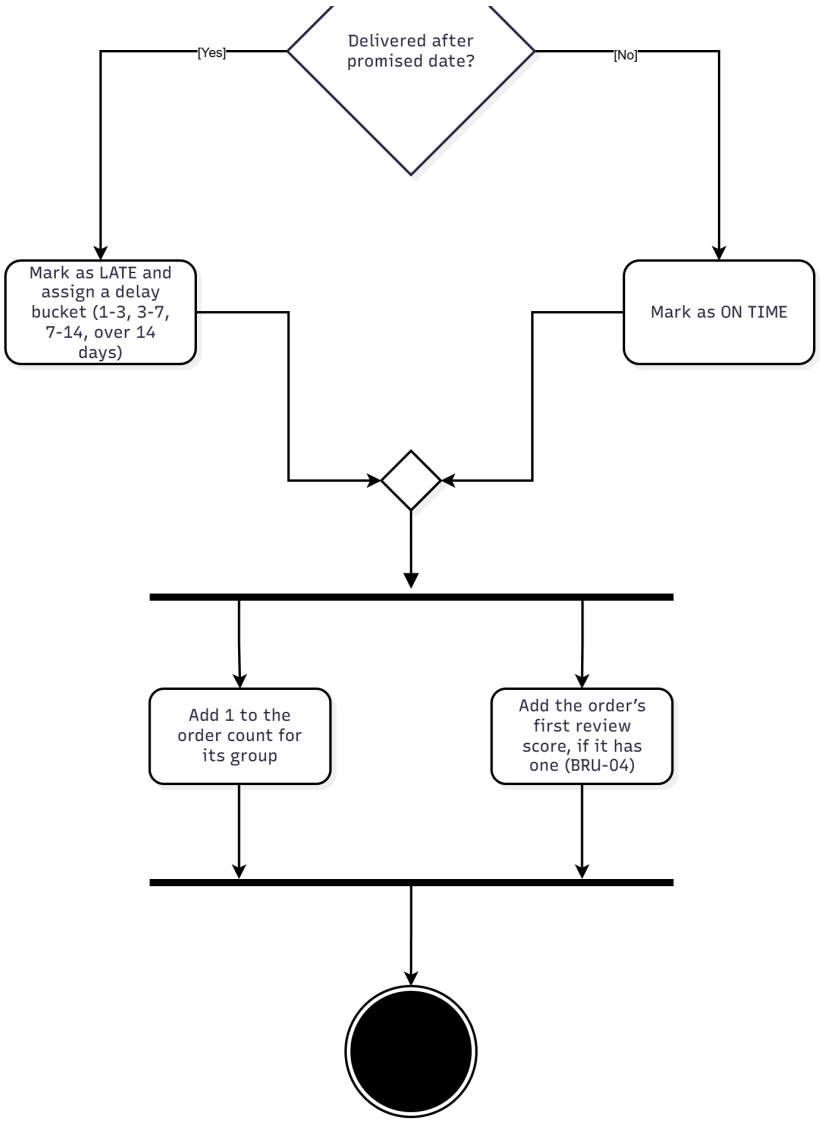


Figure 3. How each order is classified (03_activity_diagram.png, a UML activity diagram drawn in draw.io).

5. RACI Matrix

Each activity has exactly one accountable role (A). R = responsible, C = consulted, I = informed, and a dash means not involved. A/R means the same role does the work and answers for it. The roles are assumed, because this is a practice case study.

Activity	Head of Operations	Logistics Partner Manager	Seller Management Lead	Customer Experience Manager	Commercial and Finance Analyst	Data and BI Team	Business Analyst
1. Approve KPI definitions and targets	A	C	C	C	C	C	R
2. Refresh source data and run data checks	I	I	I	I	I	A/R	C
3. Build and publish the dashboard	I	I	I	I	I	A	R
4. Review monthly delivery performance	A/R	R	R	R	C	C	C
5. Investigate regions and carriers when the target is missed	I	A/R	-	-	-	C	C
6. Follow up with sellers above the late-rate threshold	I	C	A/R	I	-	C	I
7. Review customer impact of severely late orders	I	C	C	A/R	-	-	C
8. Confirm revenue figures used in reporting	I	-	-	-	A/R	C	C

6. Assumptions

No one was interviewed, so the processes here are worked out from the data and proposed by me, not observed.

- The AS-IS steps are the order journey visible in the data (purchase, approval, carrier pickup, delivery, review), not a record of internal operations.
- The absence of any routine delivery monitoring (PP-01) is the scenario set in the BRD.
- The TO-BE process, the 5% late-rate target and the monthly review cycle are proposals to be confirmed with stakeholders.
- Roles in the RACI are assumed (see BRD section 5).
- Not every view proposed in the TO-BE process was built. The gaps are listed in section 8 of the Requirements Specification.

How this document was prepared

This Process Analysis was prepared with the assistance of Claude, an AI assistant from Anthropic. Claude made the documentation process smoother by suggesting the industry-standard structure, helping with the wording, and checking that the numbers match across the Excel, SQL and Power BI work. It also answered my questions and pointed out my mistakes. The data analysis in Excel, SQL and Power BI and the three diagrams, drawn in draw.io, were done by me. Claude helped me with some of the SQL queries, which I ran and checked against my Excel results. The roles, processes and targets are assumed, because this is a practice case study.